

EE708: Assignment 4

Q1. Gini Index

Definition:

$$G = \sum_k p_k(1 - p_k) = 1 - \sum_k p_k^2$$

(a) Before Split

Given a dataset of $N = 200$ samples with 120 positive (n_+) and 80 negative (n_-) samples[cite: 4]. The probabilities are:

$$p_+ = \frac{120}{200} = 0.6, \quad p_- = \frac{80}{200} = 0.4$$

The initial Gini index is:

$$G_{initial} = 1 - (p_+^2 + p_-^2) = 1 - (0.6^2 + 0.4^2) = 1 - (0.36 + 0.16) = \mathbf{0.48}$$

(b) After Split

The split results in two subsets[cite: 5]:

- **Left Subset** ($N_L = 60$): 50 positive, 10 negative[cite: 6].

$$G_L = 1 - \left[\left(\frac{50}{60} \right)^2 + \left(\frac{10}{60} \right)^2 \right] = 1 - \left[\frac{25}{36} + \frac{1}{36} \right] = 1 - \frac{26}{36} \approx \mathbf{0.2778}$$

- **Right Subset** ($N_R = 140$): 70 positive, 70 negative[cite: 7].

$$G_R = 1 - \left[\left(\frac{70}{140} \right)^2 + \left(\frac{70}{140} \right)^2 \right] = 1 - [0.5^2 + 0.5^2] = \mathbf{0.5}$$

The weighted Gini index for the split is[cite: 8]:

$$G_{split} = \frac{N_L}{N} G_L + \frac{N_R}{N} G_R = \frac{60}{200}(0.2778) + \frac{140}{200}(0.5)$$
$$G_{split} = 0.3(0.2778) + 0.7(0.5) = 0.0833 + 0.35 = \mathbf{0.4333}$$

Conclusion: Since $0.4333 < 0.48$, the split improves purity.

Interpretation:

$$G_{split} < G_{before} \Rightarrow \text{purity increases}$$

Q2. Regression Tree (SSE)

(a) Determining best split

We compute SSE for all possible splits.

All splits:

Split	$\bar{y}_L$	$\bar{y}_R$	SSE
1.5	10	21.29	271.43
2.5	11	22.83	151.33
3.5	12.33	24.4	107.33
4.5	13.75	26	82.74
5.5	15.2	27.67	94.01
6.5	16.83	29	122.0
7.5	18.43	30	189.0

Best split at $x_1 = 4.5$

Left:

$$\begin{aligned}
 y_L &= \{10, 12, 15, 18\} \\
 \bar{y}_L &= \frac{10 + 12 + 15 + 18}{4} = \frac{55}{4} = 13.75 \\
 SSE_L &= (10 - 13.75)^2 + (12 - 13.75)^2 + (15 - 13.75)^2 + (18 - 13.75)^2 \\
 &= 14.06 + 3.06 + 1.56 + 18.06 = 36.74
 \end{aligned}$$

— Right:

$$\begin{aligned}
 y_R &= \{21, 25, 28, 30\} \\
 \bar{y}_R &= \frac{21 + 25 + 28 + 30}{4} = 26 \\
 SSE_R &= (21 - 26)^2 + (25 - 26)^2 + (28 - 26)^2 + (30 - 26)^2 \\
 &= 25 + 1 + 4 + 16 = 46
 \end{aligned}$$

— Total:

$$SSE = 36.74 + 46 = 82.74$$

(b) First split of regression tree

The dataset is split as:

$$x_1 < 4.5 \quad \text{and} \quad x_1 \geq 4.5$$

Predictions:

$$\hat{y}_L = 13.75, \quad \hat{y}_R = 26$$

Thus, the first split minimizes SSE and forms the root of the regression tree.

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Q3. Vector Quantization Distortion Analysis

(a) Distance Comparisons and New Centroids

Before assigning points, we compute the squared Euclidean distance, $d^2(x, C) = (x_1 - c_1)^2 + (x_2 - c_2)^2$, from each point to all initial centroids: $C_1(2, 3)$, $C_2(5, 8)$, and $C_3(9, 4)$.

- **Point (1,2):**

- $\rightarrow C_1: (1 - 2)^2 + (2 - 3)^2 = 1 + 1 = 2$ (**Min**)

- $\rightarrow C_2: (1 - 5)^2 + (2 - 8)^2 = 16 + 36 = 52$

- $\rightarrow C_3: (1 - 9)^2 + (2 - 4)^2 = 64 + 4 = 68$

- **Point (3,4):**

- $\rightarrow C_1: (3 - 2)^2 + (4 - 3)^2 = 1 + 1 = 2$ (**Min**)

- $\rightarrow C_2: (3 - 5)^2 + (4 - 8)^2 = 4 + 16 = 20$

- $\rightarrow C_3: (3 - 9)^2 + (4 - 4)^2 = 36 + 0 = 36$

- **Point (6,7):**

- $\rightarrow C_1: (6 - 2)^2 + (7 - 3)^2 = 16 + 16 = 32$

- $\rightarrow C_2: (6 - 5)^2 + (7 - 8)^2 = 1 + 1 = 2$ (**Min**)

- $\rightarrow C_3: (6 - 9)^2 + (7 - 4)^2 = 9 + 9 = 18$

- **Point (5,5):** (Explicitly showing why it goes to C_2)

- $\rightarrow C_1: (5 - 2)^2 + (5 - 3)^2 = 9 + 4 = 13$

- $\rightarrow C_2: (5 - 5)^2 + (5 - 8)^2 = 0 + 9 = 9$ (**Min**)

- $\rightarrow C_3: (5 - 9)^2 + (5 - 4)^2 = 16 + 1 = 17$

- **Point (8,3):**

- $\rightarrow C_1: (8 - 2)^2 + (3 - 3)^2 = 36 + 0 = 36$

- $\rightarrow C_2: (8 - 5)^2 + (3 - 8)^2 = 9 + 25 = 34$

- $\rightarrow C_3: (8 - 9)^2 + (3 - 4)^2 = 1 + 1 = 2$ (**Min**)

Based on these distance assignments, the new centroids are:

- C_1^{new} : Assigned points (1,2) and (3,4).

$$C_1^{new} = \left(\frac{1+3}{2}, \frac{2+4}{2} \right) = (2, 3)$$

- C_2^{new} : Assigned points (6,7) and (5,5).

$$C_2^{new} = \left(\frac{6+5}{2}, \frac{7+5}{2} \right) = (5.5, 6)$$

- C_3^{new} : Assigned point (8,3).

$$C_3^{new} = (8, 3)$$

(b) Distortion Calculation

The distortion function is the sum of squared Euclidean distances.

Initial Distortion (D_{init}):

Using the initial distances calculated above for assigned clusters:

$$D_{init} = 2 + 2 + 2 + 9 + 2 = 17$$

Final Distortion (D_{final}):

Using updated centroids:

- C_1 : $d^2((1, 2), (2, 3)) + d^2((3, 4), (2, 3)) = 2 + 2 = 4$
- C_2 : $d^2((6, 7), (5.5, 6)) + d^2((5, 5), (5.5, 6)) = 1.25 + 1.25 = 2.5$
- C_3 : $d^2((8, 3), (8, 3)) = 0$

$$D_{final} = 4 + 2.5 + 0 = 6.5$$

Conclusion: Since $6.5 < 17$

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Q4. Maximizing Likelihood for Σ_k and π_k

The expectation of the complete-data log-likelihood is given by:

$$E_Z[\ln p(X, Z|\mu, \Sigma, \pi)] = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk})(\ln \pi_k + \ln \mathcal{N}(x_n|\mu_k, \Sigma_k))$$

Responsibilities $\gamma(z_{nk})$ are fixed, $\therefore$ its constant.

From the Gaussian density function,

$$\ln \mathcal{N}(x_n|\mu_k, \Sigma_k) = -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k)$$

where d = dimension of data, as:

$$\mathcal{N}(x_n|\mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2}} \frac{1}{|\Sigma_k|^{1/2}} \exp \left\{ -\frac{1}{2} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k) \right\}$$

Substituting into log likelihood fn:

$$E_Z[\ln p(X, Z|\mu, \Sigma, \pi)] = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) \left[\ln \pi_k - \frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k) \right] + C$$

where C is a Constant independent of Σ_k, π_k .

(i) Maximizing the expectation value with respect to Σ_k

To differentiate with respect to the covariance matrix Σ_k , we rely on two key matrix calculus identities (assuming Σ_k is symmetric):

- **Derivative of the log determinant:** $\frac{\partial \ln |\Sigma_k|}{\partial \Sigma_k} = \Sigma_k^{-1}$
- **Derivative of the quadratic form:** $\frac{\partial}{\partial \Sigma_k} (a^T \Sigma_k^{-1} a) = -\Sigma_k^{-1} a a^T \Sigma_k^{-1}$

Applying these identities to our expectation function:

$$\frac{\partial}{\partial \Sigma_k} \sum_{n=1}^N \gamma(z_{nk}) \left[-\frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k) \right] = 0$$

Substituting the identities:

$$-\frac{1}{2} \sum_{n=1}^N \gamma(z_{nk}) [\Sigma_k^{-1} - \Sigma_k^{-1} (x_n - \mu_k)(x_n - \mu_k)^T \Sigma_k^{-1}] = 0$$

Multiply both sides by Σ_k on the left and right (i.e., $\Sigma_k[\dots]\Sigma_k$):

$$\sum_{n=1}^N \gamma(z_{nk}) [\Sigma_k - (x_n - \mu_k)(x_n - \mu_k)^T] = 0$$

Rearranging the terms:

$$\Sigma_k \sum_{n=1}^N \gamma(z_{nk}) = \sum_{n=1}^N \gamma(z_{nk})(x_n - \mu_k)(x_n - \mu_k)^T$$

Since $N_k = \sum_{n=1}^N \gamma(z_{nk})$, we isolate Σ_k :

$$\Sigma_k = \frac{\sum_{n=1}^N \gamma(z_{nk})(x_n - \mu_k)(x_n - \mu_k)^T}{N_k}$$

(ii) Maximize the expectation wrt π_k

Term involving π_k is:

$$\sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) \ln \pi_k$$

Adding the constraint ($\sum_{k=1}^K \pi_k = 1$) Using Lagrange Multiplier Method:

$$\mathcal{L} = \sum_{k=1}^K \sum_{n=1}^N \gamma(z_{nk}) \ln \pi_k + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

Differentiating w.r.t. π_k :

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = 0 \implies \sum_{n=1}^N \gamma(z_{nk}) \frac{1}{\pi_k} + \lambda = 0 \quad \dots (1)$$

Summing over all k ,

$$\begin{aligned} \sum_{k=1}^K \pi_k &= -\frac{1}{\lambda} \sum_{k=1}^K \sum_{n=1}^N \gamma(z_{nk}) \\ 1 &= -\frac{1}{\lambda} N \implies \lambda = -N \end{aligned}$$

Put in (1):

$$\begin{aligned} \pi_k &= -\frac{1}{\lambda} \sum_{n=1}^N \gamma(z_{nk}) \\ \pi_k &= \frac{N_k}{N} \quad \left(\because \sum_{n=1}^N \gamma(z_{nk}) = N_k \right) \end{aligned}$$

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Q5. Conditional Mixture Distribution

Probability Density Function (PDF) of the mixture Distribution is given by:

$$p(x) = \sum_{k=1}^K \pi_k p(x|k)$$

π_k = Mixing coefficients

$p(x|k)$ = component density of the Mixture Model for component k

now, vector x is partitioned into two parts, $x = (x_a, x_b)$.

The conditional probability density function is given by,

$$p(x_b|x_a) = \frac{p(x_a, x_b)}{p(x_a)}$$

Since $p(x)$ is a Mixture Distribution,

$$p(x_a, x_b) = p(x) = \sum_{k=1}^K \pi_k p(x|k) = \sum_{k=1}^K \pi_k p(x_a, x_b|k)$$

$$p(x_a) = \int p(x_a, x_b) dx_b = \sum_{k=1}^K \pi_k p(x_a|k)$$

where $p(x_a|k) = \int p(x_a, x_b|k) dx_b$

$$\therefore p(x_b|x_a) = \frac{\sum_{k=1}^K \pi_k p(x_a, x_b|k)}{\sum_{k=1}^K \pi_k p(x_a|k)}$$

Rewriting, we get:

$$p(x_b|x_a) = \sum_{k=1}^K \frac{\pi_k p(x_a|k)}{p(x_a)} p(x_b|x_a, k)$$

$$\bar{\pi}_k = \frac{\pi_k p(x_a|k)}{p(x_a)} \quad \text{are updated Mixing coefficients}$$

$$p(x_b|x_a, k) = \frac{p(x_a, x_b|k)}{p(x_a|k)} \quad \text{are component conditional densities}$$

$$\therefore p(x_b|x_a) = \sum_{k=1}^K \bar{\pi}_k p(x_b|x_a, k)$$

We see $p(x_b|x_a)$ as a Mixture Distribution with new Mixing components $\bar{\pi}_k$ and component density $p(x_b|x_a, k)$.

Q6. MLE Update Rules (Known Assignments)

(a) Complete Log-Likelihood & Role of z_{nk}

For a dataset $X = \{x_1, \dots, x_N\}$, the incomplete log-likelihood is:

$$\mathcal{L}(\theta) = \sum_{n=1}^N \log \sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)$$

Why z_{nk} simplifies this: The inner summation couples all parameters, preventing analytical derivatives. By introducing the known assignment (one-hot indicator variable) z_{nk} , where $z_{nk} = 1$ if point n belongs to cluster k and 0 otherwise, we completely remove the inner summation. Only one term in the product survives, turning the logarithm of a sum into a sum of logarithms.

$$\mathcal{L}_c(\theta) = \sum_{n=1}^N \sum_{k=1}^K z_{nk} [\log \pi_k + \log \mathcal{N}(x_n | \mu_k, \Sigma_k)]$$

(b) MLE Update Rules

Intuition: Hard assignments (z_{nk}) decouple the model into K independent Gaussian estimation problems. We estimate parameters for each cluster using only its assigned points, weighted by z_{nk} . Let $N_k = \sum_{n=1}^N z_{nk}$ be the number of points in cluster k .

1. Update Rule for π_k (Mixing Coefficients)

Isolate π_k terms and add the constraint $\sum_{k=1}^K \pi_k = 1$ using a Lagrange multiplier λ :

$$\mathcal{L}(\pi, \lambda) = \sum_{n=1}^N \sum_{k=1}^K z_{nk} \log \pi_k + \lambda \left(1 - \sum_{k=1}^K \pi_k \right)$$

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{n=1}^N z_{nk}}{\pi_k} - \lambda = 0 \implies \pi_k = \frac{N_k}{\lambda}$$

Summing over k : $\sum_{k=1}^K \pi_k = \frac{1}{\lambda} \sum_{k=1}^K N_k \implies 1 = \frac{N}{\lambda} \implies \lambda = N$.

$$\therefore \pi_k = \frac{N_k}{N}$$

This explicitly shows that π_k is the fraction of total points assigned to cluster k .

2. Update Rule for μ_k (Mean)

Isolate μ_k terms and differentiate. Using $\frac{\partial}{\partial \mu} [(x - \mu)^T A (x - \mu)] = -2A(x - \mu)$:

$$J(\mu_k) = -\frac{1}{2} \sum_{n=1}^N z_{nk} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k)$$

$$\frac{\partial J}{\partial \mu_k} = \sum_{n=1}^N z_{nk} \Sigma_k^{-1} (x_n - \mu_k) = 0$$

Multiply by Σ_k to remove the inverse:

$$\sum_{n=1}^N z_{nk} (x_n - \mu_k) = 0 \implies \sum_{n=1}^N z_{nk} x_n = \mu_k \sum_{n=1}^N z_{nk}$$

$$\therefore \mu_k = \frac{1}{N_k} \sum_{n=1}^N z_{nk} x_n$$

3. Update Rule for Σ_k (Covariance Matrix)

Isolate Σ_k terms and differentiate. Using $\frac{\partial \ln |\Sigma|}{\partial \Sigma} = \Sigma^{-1}$ and $\frac{\partial a^T \Sigma^{-1} a}{\partial \Sigma} = -\Sigma^{-1} a a^T \Sigma^{-1}$:

$$J(\Sigma_k) = -\frac{1}{2} \sum_{n=1}^N z_{nk} [\ln |\Sigma_k| + (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k)]$$

$$\frac{\partial J}{\partial \Sigma_k} = -\frac{1}{2} \sum_{n=1}^N z_{nk} [\Sigma_k^{-1} - \Sigma_k^{-1} (x_n - \mu_k)(x_n - \mu_k)^T \Sigma_k^{-1}] = 0$$

Multiply left and right by Σ_k :

$$\sum_{n=1}^N z_{nk} [\Sigma_k - (x_n - \mu_k)(x_n - \mu_k)^T] = 0$$

$$\Sigma_k \sum_{n=1}^N z_{nk} = \sum_{n=1}^N z_{nk} (x_n - \mu_k)(x_n - \mu_k)^T$$

$$\therefore \Sigma_k = \frac{1}{N_k} \sum_{n=1}^N z_{nk} (x_n - \mu_k)(x_n - \mu_k)^T$$

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